

Encoding Nothing at the Edge of Stability: Missing Electrodes, Coordinate Origins, and Empty Baselines in Spiking Reservoir Representations of Affective EEG

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Abstract—ABSTRACT-PLACEHOLDER

Index Terms—EEG, event-related potentials, spiking neural networks, reservoir computing, affective computing, missing data, normalization, neuromorphic computing.

I. INTRODUCTION

Electroencephalographic event-related potentials (ERPs) measure evoked neural dynamics at millisecond resolution, but the recorded waveform is a noisy, low-dimensional, and subject-dependent projection of an unobserved neural process [14], [15]. A representation of affective ERPs can separate conditions under one protocol and keep little usable structure after amplitude noise, temporal misalignment, or electrode loss, and endpoint accuracy alone does not say which. Spiking reservoirs make the gap sharper: fixed recurrent networks give nonlinear temporal expansions [1], [4], yet EEG studies usually score them as classifiers [11], [17] rather than as measurement operators whose invariances can be tested.

Every such test has to decide what *nothing* means. An absent electrode must be represented by some number. A feature coordinate has an origin, and a zero-filled block sits at that origin whether or not it means anything there. The interval before stimulus onset contains no stimulus-locked signal, yet it is part of every epoch the encoder reads. A score has to be compared with the floor set by no information at all. And a reservoir’s recurrence can be switched off ($\rho = 0$) to ask what the dynamics add. These choices are rarely stated, and each one can move a measured result.

Our conference study [30] measured two of them on one cohort. On SHAPE (211 subjects), raw retention rewarded the encoder closest to chance, and zero-filling a dropped channel before standardization, which is not invariant to re-centering the features (Proposition 1), halved the measured lead of ERP-window features over a near-centered spiking-reservoir

code at 30% dropout (+0.049 versus +0.092 to +0.107 BA); on DEAP the fill rule alone moved band-power BA by up to 0.11. The reservoir, a fixed leaky integrate-and-fire (LIF) network read out through six spike-count bins (BSC_6), sat just past its measured edge of stability ($\rho^* \approx 0.82$, below the echo-state heuristic $\rho = 1$), and removing its recurrence cost 0.049 BA. That study left five questions open. It used one cohort, one reservoir draw, and one EEGNet recipe. It argued Proposition 1 algebraically but never moved the origin to watch the score follow. And a post-publication code check found that its reservoir window, $t \in [10, 70)$ samples, spans about -160 to $+73$ ms around onset, so most of what the reservoir encoded came before the stimulus.

This paper answers those questions with a second, public cohort and new experiments. The contributions are:

- 1) An external, paradigm-matched test on an independent IAPS cohort (228 subjects, different laboratory, montage, and amplifier) at the conference-matched epoch and at a long epoch pre-specified from the dataset’s published effect latencies (Secs. IV, V-A).
- 2) A direct experimental test of Proposition 1: shifting the coordinate origin leaves every clean prediction unchanged while the dropout score follows the standardized position of the fill (Sec. V-C).
- 3) Proposition 2, which shows that per-epoch z-scoring writes a function of the post-stimulus signal into the pre-stimulus interval, and a test of how much of each encoder’s accuracy that interval carries (Secs. III-G, V-D).
- 4) The operating point across ten reservoir draws, thresholds, and leaks: where the measured edge of stability lies and whether accuracy follows it (Sec. V-E).
- 5) Two physically grounded perturbations, spherical-spline repair of removed electrodes and re-averaging from fewer recorded trials, together with EEGNet with and without augmentation and across recipes on the second cohort (Sec. V-G).

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A preliminary version of this work was accepted at the 2026 IEEE International Conference on Bioinformatics and Biomedicine (BIBM) [30]. This version adds an independent public cohort, a direct test of Proposition 1, Proposition 2 and the pre-stimulus experiments, the operating point across ten reservoir draws, spherical-spline repair and recorded trial-count noise, and EEGNet recipe variation.

The affective categories define the task. The object of study is what each representation does with nothing: with a missing electrode, a shifted origin, an empty baseline, and a chance floor.

II. RELATED WORK

ERP representations of affect. Emotional pictures enlarge the late positive potential relative to neutral ones from roughly 400 ms after onset [43], [46], [47], and a recent cohort of 224 participants adds an ordered valence effect between 1500 and 2500 ms [32]. ERP/EEG decoders commonly use window amplitudes, Hjorth [12] and spectral descriptors [13], and linear models [15], [44], which are interpretable through these known components but can miss nonlinear temporal structure. Deep networks such as EEGNet [16] are now common [17], [45], and much of this literature reports endpoint accuracy over preprocessing and acquisition choices [20].

Reservoirs and the edge of stability. Reservoir computing maps an input through a fixed high-dimensional recurrent system to a state trajectory read out by a simple classifier [1]–[4], and spiking liquid-state machines add event timing and count structure [7], [8]. Computation is often reported to peak near the transition between ordered and irregular dynamics, located by how far a small state perturbation spreads [5], [6], [34]. The spectral radius $\rho = 1$ is a heuristic for rate reservoirs, and it is neither necessary nor sufficient for the echo-state property of a driven system [35]; for a thresholded spiking reservoir the transition need not lie at $\rho = 1$. Neuromorphic reservoir features for affective EEG were introduced in earlier ARSPI-Net work [9], [10], and this paper extends our conference study [30].

Missing electrodes. Dynamic spatial filtering handles missing channels with a learned attention front end [23], and channel-dropout or noise augmentation hardens trained models [24]. A recent EEG foundation-model benchmark finds that apparent channel-dropout fragility depends on whether dropped channels are removed or zero-padded [22]. Fills range from constants and spherical-spline interpolation [25], through learned imputation [23], to generative synthesis of whole missing modalities [26], [27]. The non-learned fills studied here sit at the simple end of that spectrum.

Evaluation choices that create or remove effects. High-pass filters can move late ERP effects into earlier latencies, including the pre-stimulus baseline [38]–[40], and leakage between preprocessing and evaluation inflates reported accuracy across machine-learning science [41], [42]. Decoding accuracy near a theoretical chance level needs empirical null distributions [36], and cross-validated accuracy on a few hundred observations carries wide error bars [37]. Per-epoch normalization, the fill convention for an absent electrode, and the origin of a feature coordinate belong to the same family of choices; we give each a proposition or a direct test.

III. METHODS

A. ERP Observation Model

Let $X_s^{(c)} \in \mathbb{R}^{C \times T}$ denote the trial-averaged ERP observation of subject s in condition c . We treat it as a noisy readout of an unobserved neural state trajectory,

$$\xi_{t+1} = f_\theta(\xi_t, u_t) + \eta_t, \quad x_t = H\xi_t + \epsilon_t, \quad (1)$$

where H is the scalp measurement operator and ϵ_t collects measurement noise, residual single-trial variability, and preprocessing. Every perturbation below acts on a term of this

model: electrode removal zeroes rows of H , additive noise and re-averaging from fewer trials enlarge ϵ_t , and alignment jitter shifts the sampling index of x_t . Normalization acts on x_t before any encoder sees it, which is why it is treated as part of the measurement (Sec. III-G).

B. Spiking Reservoir Encoding

Each electrode trace drives a fixed LIF reservoir of 256 units with input weights W_{in} , recurrent weights W , leak $\beta = 0.05$, and threshold $\vartheta = 0.5$:

$$v_t = [(1 - \beta)(1 - z_{t-1}) \odot v_{t-1} + Wz_{t-1} + W_{\text{in}}x_t]_+, \quad (2)$$

$$z_t = \mathbf{1}[v_t \geq \vartheta], \quad (3)$$

where v_t is the membrane potential, z_t the binary spike vector, and $[\cdot]_+$ rectification at zero; the factor $(1 - z_{t-1})$ resets every unit that spiked at the previous step. W_{in} and W are drawn once from a Xavier-uniform law, and W is rescaled to spectral radius ρ . Unless stated otherwise the draw is seed 42 and $\rho = 0.9$, the conference operating point; only the readout is trained.

Activity is summarized by six equal temporal bins over an analysis window. For unit j and bin k , the binned spike-count code is $b_{j,k} = \sum_{t \in \mathcal{B}_k} z_j(t)$. Each electrode's 1536-dimensional spike-bin vector is projected onto a single 64-component PCA basis shared across electrodes and fitted on the training subjects of each fold, and the projections are concatenated, giving $E \in \mathbb{R}^{64C}$ (2176 for SHAPE, 2048 for the external cohort):

$$E = \phi_{\text{PCA}}([b_1, \dots, b_6]) = \Phi(X). \quad (4)$$

The conference window is $t \in [10, 70]$ in samples. With onset at sample 51 in both cohorts it spans about -160 to $+73$ ms around onset, so most of it precedes the stimulus. Section III-G examines what that window can read, and Sec. V-D adds windows wholly before and wholly after onset.

A silent electrode is a fixed point of the reservoir: with $x_t = 0$ from a zero initial state, v_t stays at zero and no unit spikes, so the electrode's spike-bin block is exactly zero. After the PCA step that block maps to $-P\bar{b}$, the negative of the projected training mean, not to the origin of the embedding.

C. Operating Point and Order Parameter

To test whether the recurrent dynamics matter, the same draw is rescaled to $\rho \in \{0, 0.3, 0.6, 0.8, 0.9, 1.0, 1.2, 1.5\}$ with every other parameter fixed ($\rho = 0$ leaves a feedforward LIF layer). A label-free order parameter locates the dynamical transition by damage spreading [5]: two copies of the reservoir receive the same electrode drive, one unit's spike is flipped at $t = 10$, and the normalized Hamming distance between the copies' spike vectors is averaged over steps 21 to 69 and over 400 sampled drives and flipped units. The transition ρ^* is the ρ at which this damage, linearly interpolated on the grid $\rho = 0, 0.1, \dots, 2$, reaches half its maximum. The conference measured ρ^* for one draw on SHAPE drives. Here the order parameter and the accuracy sweep are repeated for ten independent draws (seeds 42–51) on the external drives, and ρ^* is also measured at thresholds $\vartheta \in \{0.25, 0.5, 1.0\}$ and leaks $\beta \in \{0.02, 0.05, 0.1, 0.2\}$.

D. Baselines

Band-power uses Welch power in five bands (1–4, 4–8, 8–13, 13–30, 30–45 Hz) per channel (5C dimensions). ERP-window uses mean amplitude in three windows, about –120–0 ms, 50–250 ms, and 250–600 ms relative to onset, per channel (3C dimensions). All three fixed encoders feed a balanced L2-regularized logistic readout. As a trained baseline, EEGNet-8,2 [16] is trained per fold on the raw $C \times T$ signal (Adam, learning rate 10^{-3} , 80 epochs, batch 64, dropout 0.25) with train-only per-channel standardization, with and without augmentation by random channel dropout and additive noise. On the external cohort the recipe is also varied (40 or 160 epochs, dropout 0.5) to bound the effect of a single recipe.

E. Perturbations

The feature-coordinate experiment suppresses a dropped channel’s feature block after extraction and refills it by one of four rules: zero, train mean, train-fitted k -nearest-neighbor imputation ($k=10$), and the mean of the retained channels (spatial). It isolates the fill convention and applies to the fixed encoders.

The signal-level experiment corrupts the test subjects’ signal before any feature extraction, identically for every encoder including EEGNet, with the same draws in each fold, while training uses the clean signal: removal of 10, 30, or 50% of the electrodes (a removed electrode is a zero trace), additive white noise at 5 dB SNR per channel, and a uniform alignment shift of up to ± 50 ms. Two signal-level conditions are new. *Spline repair* replaces each removed electrode by spherical-spline interpolation from the retained electrodes [25] (order 4, seven Legendre terms) on the baseline-corrected microvolt signal before normalization, a physically grounded alternative to the zero trace. *Recorded observation noise* re-averages every test observation from 1, 4, or 16 of its own single trials instead of all of them, so the added variance is the recorded trial-to-trial variability rather than synthetic noise; it needs single trials and is run on the external cohort only.

F. Retention and the Coordinate Origin

Raw retention, the ratio of perturbed to clean BA, overstates the stability of representations whose clean BA is close to chance ($c = 1/3$), so we also report *above-chance retention*, $(\text{BA}_{\text{pert}} - c)/(\text{BA}_{\text{clean}} - c)$, and treat absolute BA and paired subject-level differences as primary.

Write the standardized readout input as $z = D_{\sigma}^{-1}(x - \mu)$ with $D_{\sigma} = \text{diag}(\sigma)$, and let S index the dropped coordinates. Zero-fill sets $x_S = 0$, so the dropped block enters as $z_S = -D_{\sigma,S}^{-1}\mu_S$; train-mean fill sets $x_S = \mu_S$, so $z_S = 0$:

$$z_S^{\text{zero}} - z_S^{\text{mean}} = -D_{\sigma,S}^{-1}\mu_S. \quad (5)$$

Proposition 1 (origin dependence of zero-fill). *Let the clean readout depend on x only through z , so it is invariant to a constant offset $x \mapsto x - b$ (train-only standardization guarantees this). Then the clean prediction is unchanged by the re-centering, while the zero-fill dropout input shifts by $\Delta z_S = D_{\sigma,S}^{-1}b_S$. Hence the measured zero-fill dropout*

accuracy depends on the coordinate origin, and the zero- and mean-fill inputs coincide if and only if $\mu_S = 0$.

Proof. Standardization subtracts the training mean, so b cancels from the clean input. A zero-filled coordinate maps to $z_i = -(\mu_i - b_i)/\sigma_i$ instead of $-\mu_i/\sigma_i$, a shift of $b_i/\sigma_i \neq 0$ for any $b_i \neq 0$. $\square$

For a linear readout with class weights W , Eq. (5) moves the class logits by $\Delta \ell = -W_S D_{\sigma,S}^{-1}\mu_S$ between the two fills, so the fills can disagree only as far as the dropped features are non-centered. Proposition 1 has a direct experimental test, run here for the first time: fill the dropped block with the raw value $x_i = \mu_i + \kappa\sigma_i$, which places it at standardized position $z_i = \kappa$ for every encoder, and sweep κ . Clean predictions must not change with any offset b , while dropout accuracy must trace a curve in κ on which train-mean fill is $\kappa = 0$ and each encoder’s own zero sits near $\kappa = -|\mu|/\sigma$. We refit the readout on features shifted by $b = \mu + 2\sigma$ to check the first prediction and sweep $\kappa \in [-3, 3]$ to test the second.

G. What the Pre-Stimulus Interval Can Contain

Both cohorts are baseline-corrected over the 200 ms before onset, so the pre-stimulus mean of every trial-averaged trace is zero, and both are then z-scored per channel over the whole epoch, as in the conference pipeline. Let P be the pre-stimulus samples and Q the rest, with $|P| + |Q| = T$, and let m_Q be the post-stimulus mean and s the epoch standard deviation of one channel.

Proposition 2 (baseline transfer under per-epoch normalization). *For a baseline-corrected trace normalized by its own epoch mean and standard deviation, the mean of the normalized pre-stimulus segment is $-\frac{|Q|}{T}m_Q/s$. The pre-stimulus segment is therefore a deterministic function of the post-stimulus signal, and any encoder that reads only P has access to m_Q/s .*

Proof. The epoch mean is $(|P| \cdot 0 + |Q|m_Q)/T$. Subtracting it and dividing by s maps the zero pre-stimulus mean to $-|Q|m_Q/(Ts)$. $\square$

With onset at sample 51 of 256, $|Q|/T = 0.80$. An affective modulation of the late positive potential changes m_Q , so under per-epoch z-scoring a classifier can separate conditions from an interval in which no stimulus has yet been shown. Train-only per-channel scaling (one scale per channel from the training subjects, applied to every epoch) leaves the pre-stimulus mean at zero and removes this route. We compare the two normalizations for windows wholly before onset, the conference window, and windows wholly after onset (Sec. V-D).

IV. DATA AND PROTOCOL

A. SHAPE Cohort

The primary cohort is the trial-averaged affective ERP data of the Stress, Health, and the Psychophysiology of Emotion (SHAPE) project [29], collected by the Laboratory for Clinical Affective Neuroscience at Stony Brook University: 211 subjects after exclusion of one anomalous record, each contributing negative, neutral, and pleasant observations (633 in total). Each epoch (34 scalp channels at 1024 Hz, 1.2 s,

baseline-corrected over the 200 ms pre-stimulus interval) is analyzed as every fourth of its first 1024 samples: 256 samples at 256 Hz spanning -200 to $+800$ ms, onset at sample 51, z-scored per channel within each epoch. SHAPE data are held in a restricted environment; every SHAPE number in this paper is the conference result [30], regenerated from its committed code and aggregate outputs, and no SHAPE analysis was rerun for this version.

B. External Cohort

The external cohort is a public IAPS passive-viewing dataset [32], [33] from a different laboratory, montage, and amplifier: 32 scalp electrodes (BioSemi layout) referenced to the mastoids, recorded at 512 Hz, filtered, ocular-corrected with ICA, and artifact-rejected by the dataset authors. Of 239 participants, the 228 with a complete negative, neutral, and pleasant panel in the preprocessed release are used (684 observations; median 44 trials per observation, range 20 to 54; high- and low-arousal pictures of one valence are pooled). Each observation is built to the SHAPE layout: trial average, baseline correction over -200 to 0 ms, every second sample from -200 ms, giving 256 samples at 256 Hz with onset at sample 51, and z-scoring per channel within the epoch. This conference-matched epoch is the primary external analysis. The grand averages reproduce the published effects (Fig. 1): emotional pictures exceed neutral ones by $2.12 \mu\text{V}$ at PO3 between 500 and 1300 ms, and negative pictures exceed pleasant ones by $1.32 \mu\text{V}$ between 1500 and 2500 ms. Most of both effects lies after $+800$ ms, so we pre-specified, before computing any of its results, a secondary analysis on a long epoch from -200 to $+2496$ ms (691 samples, same sampling, baseline, and onset). In it the reservoir window covers 0 to $+2484$ ms after onset (six bins of 106 samples), the ERP-window encoder adds the two published windows (500–1300 and 1500–2500 ms), and band-power and EEGNet see the whole epoch.

C. DEAP Check

On DEAP [28] (32 subjects, band-power features, binary valence, within-subject 5-fold) the conference measured how much the fill rule alone moves channel-dropout accuracy. That check is reported unchanged (Fig. 6a).

D. Validation

All analyses share one fold structure: StratifiedGroupKFold (5 folds, seeds 42–46) with subjects as groups, so a subject’s observations never span training and test. Standardization and the reservoir’s PCA basis are fitted on the training subjects of each fold, the fixed encoders use a balanced L2 logistic readout, and EEGNet is trained on the same folds. The out-of-fold class probabilities of each observation are averaged over the five partitions, and 95% confidence intervals (CIs) come from 4000 resamples of the subjects with replacement, each keeping its three observations. Contrasts between encoders, fills, conditions, or operating points are paired differences under the same resampling, so no test treats the 25 overlapping

folds as independent. Chance is $1/3$, and a within-subject label-permutation null (20 permutations) gives the empirical floor for every fixed encoder [36]. The journal code reproduces the conference implementation bit for bit on synthetic data (reservoir codes, damage, and bootstrap intervals), and only aggregate results leave the restricted environment.

V. RESULTS

A. Clean Classification on Two Cohorts

Table I reports clean three-class BA, macro-F1, macro one-vs-rest ROC-AUC, and the largest BA in the permutation null for every encoder and cohort. On SHAPE, ERP-window features are the strongest fixed encoder (BA 0.629), above band-power (0.494) and the reservoir (0.487), and EEGNet is more accurate still (0.731).

The external cohort carries much less condition information in the same layout. At the conference-matched epoch only ERP-window features clear every permutation (BA 0.386 against a null maximum of 0.367); band-power sits at its null maximum (0.368 against 0.365), and the reservoir is at chance (0.320, 95% CI [0.288, 0.354]). On the pre-specified long epoch every fixed encoder clears chance: ERP-window 0.427 [0.398, 0.455], reservoir 0.396 [0.361, 0.428], and band-power 0.389 [0.360, 0.418], though only ERP-window exceeds every permutation (null maxima 0.361, 0.387, and 0.377). ERP-window leads the reservoir by $+0.030$ BA (CI $[-0.012, +0.072]$). The encoder ranking of SHAPE is therefore preserved in direction, but the external above-chance margin is about a third of SHAPE’s ($0.427 - 1/3$ versus $0.629 - 1/3$ for ERP-window), which bounds every perturbation effect measured on it. We attribute this to the cohort rather than the pipeline: the grand averages reproduce the published effects (Fig. 1), and a separate analysis of this cohort finds between-subject variance an order of magnitude larger relative to condition variance than on SHAPE [31].

The reservoir controls of the conference replicate in kind. Pooled (transductive) PCA instead of train-only PCA changes long-epoch BA by -0.006 ($[-0.018, +0.004]$), so the sensitivity check again shows no evidence of optimistic inflation from the transductive basis under this protocol. Component counts of 16, 32, 64, and 128 give 0.428, 0.434, 0.396, and 0.424, with overlapping intervals. One control changes: over the long window, six bins beat the single-bin total count under 30% electrode removal ($+0.032$ BA, CI $[+0.007, +0.056]$) and lean the same way on clean data ($+0.034$, $[-0.010, +0.076]$), whereas on SHAPE’s short window BSC_6 and BSC_1 were indistinguishable. Temporal binning carries information when the window spans effects at different latencies, here the salience effect followed by the valence effect.

B. Feature-Coordinate Fills: What Replicates

Proposition 1 names the input-level cause of the conference result: the zero-filled block sits at $-\mu_S/\sigma_S$ in standardized coordinates. That cause replicates almost exactly. The mean $|\mu|/\sigma$ of the features is 1.20, 0.75, and 0.06 for band-power, ERP-window, and the reservoir on SHAPE, and 1.20, 0.72, and 0.06 on the external long epoch (1.20, 0.68, 0.05 at

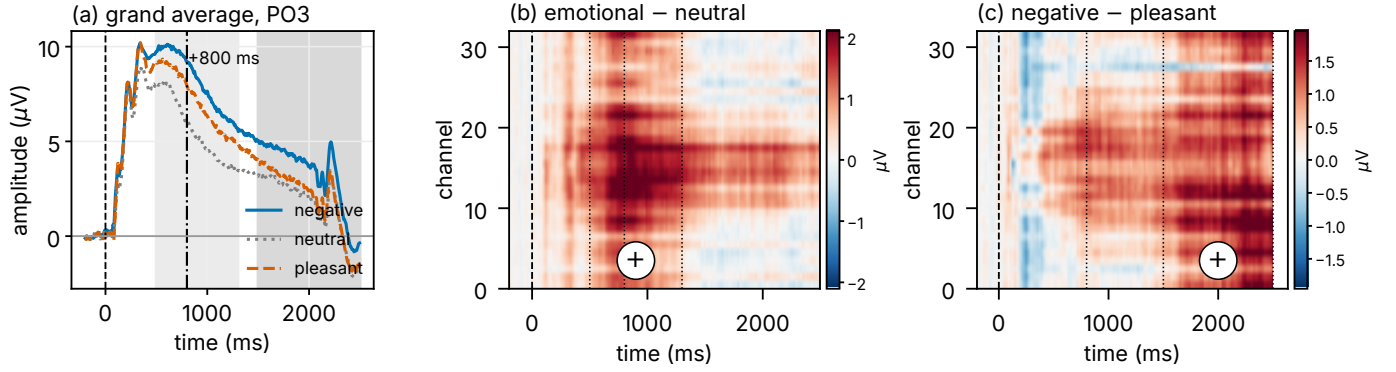


Fig. 1. Grand-average ERPs of the external cohort (228 subjects; baseline-corrected microvolts, long epoch). (a) Waveforms per valence at the channel with the largest emotional-minus-neutral difference between 500 and 1300 ms; light and dark shading mark the published salience (500–1300 ms) and valence (1500–2500 ms) windows, and the dash-dot line the end of the conference-matched epoch (+800 ms). (b) Emotional minus neutral and (c) negative minus pleasant over all 32 channels; + marks the sign of the effect in grayscale.

TABLE I

CLEAN THREE-CLASS BALANCED ACCURACY (SUBJECT-LEVEL BOOTSTRAP 95% CI), MACRO-F1, MACRO ONE-VS-REST ROC-AUC FROM POOLED OUT-OF-FOLD PROBABILITIES, AND THE LARGEST BA AMONG 20 WITHIN-SUBJECT LABEL PERMUTATIONS. SHAPE VALUES ARE THE CONFERENCE RESULTS [30]; THE CONFERENCE STORED NO EEGNET PROBABILITIES, SO F1 AND AUC ARE NOT AVAILABLE FOR IT. CHANCE IS 1/3 (AUC 0.5).

Cohort	Encoder	BA [95% CI]	F1	AUC	Null max
SHAPE ($N=211$)	ERP-window	0.629 [0.597, 0.660]	0.629	0.799	0.390
	Band-power	0.494 [0.460, 0.528]	0.492	0.682	0.378
	Reservoir	0.487 [0.447, 0.523]	0.485	0.658	0.379
	EEGNet	0.731 [0.701, 0.763]	—	—	—
	EEGNet + aug.	0.731 [0.700, 0.763]	—	—	—
External, $-200..800$ ms	ERP-window	0.386 [0.358, 0.414]	0.385	0.561	0.367
	Band-power	0.368 [0.339, 0.399]	0.367	0.531	0.365
	Reservoir	0.320 [0.288, 0.354]	0.320	0.482	0.368
External, long epoch	ERP-window	0.427 [0.398, 0.455]	0.427	0.597	0.361
	Band-power	0.389 [0.360, 0.418]	0.388	0.553	0.377
	Reservoir	0.396 [0.361, 0.428]	0.396	0.576	0.387

the matched epoch). The near-centeredness of the reservoir belongs to its PCA step, not to its spikes: the raw spike-count code is as far from centered as band-power (1.14 at the matched epoch, 2.58 on the long window), and the shared basis subtracts the pooled mean.

The accuracy consequence does not replicate at a detectable level (Table II, Fig. 2). On SHAPE, train-mean fill exceeded zero-fill at 30% dropout by +0.055 BA for band-power and +0.042 for ERP-window, and zero-fill halved the ERP-window lead over the reservoir. On the external long epoch the same contrasts are -0.013 ($[-0.038, +0.012]$) and $+0.014$ ($[-0.012, +0.039]$), and no fill moves the ERP-window lead outside noise. What does replicate is the near-centered code's stability under the fill choice: the reservoir's two constant fills agree within 0.01 BA at every rate from 10 to 50% on both external epochs, with intervals about a third as wide as those of the non-centered encoders. Section V-C shows why accuracy moved on one cohort and not on the other.

C. Moving the Origin: A Direct Test of Proposition 1

Proposition 1 makes two predictions that can be tested by moving the coordinate origin, and both hold on the external cohort (Fig. 3, Table III).

TABLE II

CHANNEL DROPOUT AT 30% (FEATURE-COORDINATE EXPERIMENT): SUBJECT-LEVEL BA UNDER FOUR FILL RULES, AND THE PAIRED DIFFERENCE ERP-WINDOW MINUS RESERVOIR UNDER THE SAME SUBJECT RESAMPLING.

Cohort	Fill	Band	ERP	Res.	ERP-Res [95% CI]
SHAPE	Zero	0.417	0.536	0.487	+0.049 [+0.003, +0.095]
	Mean	0.472	0.578	0.487	+0.092 [+0.049, +0.134]
	kNN	0.483	0.610	0.507	+0.103 [+0.062, +0.147]
	Spatial	0.488	0.589	0.482	+0.107 [+0.063, +0.150]
External, long	Zero	0.398	0.393	0.392	+0.001 $[-0.039, +0.042]$
	Mean	0.385	0.406	0.390	+0.017 $[-0.025, +0.059]$
	kNN	0.380	0.412	0.396	+0.016 $[-0.026, +0.060]$
	Spatial	0.385	0.398	0.404	-0.006 $[-0.048, +0.035]$
External, $-200..800$	Zero	0.360	0.370	0.325	+0.046 [+0.006, +0.085]
	Mean	0.360	0.354	0.325	+0.030 $[-0.012, +0.072]$
	kNN	0.376	0.389	0.319	+0.070 [+0.025, +0.113]
	Spatial	0.377	0.383	0.320	+0.063 [+0.022, +0.104]

First, clean predictions do not depend on the origin. Refitting every readout on features shifted by $b = \mu + 2\sigma$ leaves the predicted label of every test observation unchanged for every encoder and fold on both epochs; class probabilities differ by at most 5.8×10^{-3} , the stopping tolerance of the L-BFGS

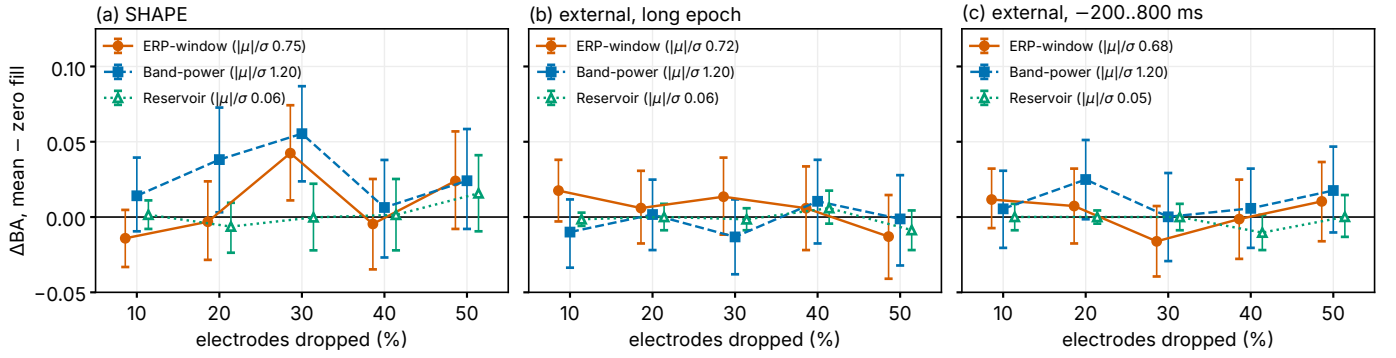


Fig. 2. How much the fill rule alone moves channel-dropout accuracy: paired subject-level change in BA (95% CI) of train-mean fill minus zero-fill against the share of electrodes dropped, for each fixed encoder; the legend gives the measured mean $|\mu|/\sigma$ of each encoder’s features on that cohort. (a) SHAPE (conference results). (b) External cohort, long epoch. (c) External cohort, conference-matched epoch. The input-level cause ($|\mu|/\sigma$) replicates on both cohorts; the accuracy change does not, because the external encoders have little accuracy above chance to lose.

solver (at most 4.2×10^{-14} for the reservoir).

Second, the dropout score is a function of where the fill sits. With 30% of the electrodes dropped on the long epoch, ERP-window BA is 0.406 with the fill at the training mean ($\kappa = 0$) and falls to 0.345 [0.330, 0.360] with the same block placed three standard deviations away ($\kappa = +3$); band-power falls from 0.385 to 0.349, and the reservoir from 0.390 to 0.349. The reservoir is not exempt: its zero sits at $\kappa \approx -0.06$, next to its training mean, so the zero-versus-mean choice is immaterial for it, but a fill far from that mean costs it as much as it costs the other encoders. Each encoder’s native zero lands where its centeredness puts it on its own curve: at $\kappa \approx -1.20$ for band-power, -0.72 for ERP-window, and -0.06 for the reservoir.

The mechanism replicates even where accuracy cannot move. Between zero-fill and mean-fill, the class logits of a test observation move by $\|\Delta\ell\| = 1.76$ for band-power, 1.45 for ERP-window, and 0.17 for the reservoir (class-centered norm, mean over folds), and the fill choice changes the predicted label of 34.5%, 32.9%, and 3.6% of the test observations, in the order of the encoders’ non-centeredness. At the conference-matched epoch, where no encoder is far from chance, the same ordering holds (37.8%, 25.9%, and 3.3% of labels change) while BA barely moves. On SHAPE the rewritten labels cost band-power 0.055 BA and halved an encoder difference; on the external cohort a similar share of rewritten labels costs little, because the encoders have about a third of SHAPE’s above-chance margin to lose. Proposition 1 thus predicts which representations a fill rule perturbs; the above-chance margin sets how much accuracy that perturbation can remove.

D. The Empty Baseline

Proposition 2 predicts that under per-epoch z-scoring the pre-stimulus mean of each normalized trace is a scaled negative of its post-stimulus mean. Across the 684×32 external traces the correlation is -0.90 at the conference-matched epoch and -0.91 on the long epoch, so the transfer happens. Whether it carries condition information depends on how much the post-stimulus mean differs between conditions (Fig. 4). ERP-window features computed only from the 200 ms before onset reach BA 0.357 [0.333, 0.382] under per-epoch

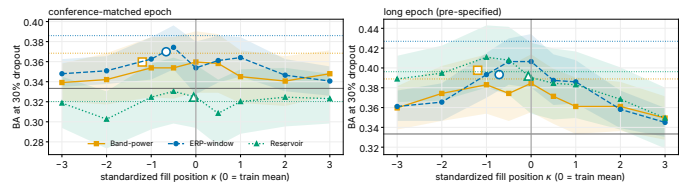


Fig. 3. Moving the fill along the standardized axis (external cohort, 30% of electrodes dropped). Subject-level BA (95% CI, shaded) versus the standardized position κ of the filled block ($\kappa = 0$: train-mean fill). Open markers: each encoder’s native zero-fill, placed at its mean $-|\mu|/\sigma$; dotted lines: clean BA, identical for every origin. (a) Conference-matched epoch, where every encoder is near chance. (b) Long epoch: moving the fill away from the training mean costs every encoder, including the reservoir.

TABLE III

ORIGIN TEST AT 30% DROPOUT (EXTERNAL COHORT, LONG EPOCH): MEAN $|\mu|/\sigma$ OF THE DROPPED BLOCK, CLEAN BA, BA WITH THE NATIVE ZERO-FILL AND WITH THE FILL AT $\kappa = -2, 0, +2$, THE CLASS-CENTERED LOGIT SHIFT BETWEEN ZERO AND MEAN FILL, AND THE PERCENTAGE OF TEST LABELS THAT THE ZERO-VERSUS-MEAN CHOICE CHANGES.

Encoder	$\frac{ \mu }{\sigma}$	Clean	Zero	$\kappa = -2$	$\kappa = 0$	$\kappa = +2$	$\ \Delta\ell\ $	Flip %
Band-power	1.20	0.389	0.398	0.374	0.385	0.361	1.76	34.5
ERP-window	0.72	0.427	0.393	0.365	0.406	0.358	1.45	32.9
Reservoir	0.06	0.396	0.392	0.395	0.390	0.368	0.17	3.6

z-scoring at the matched epoch (permutation $p = 0.075$) and 0.325 under a fixed per-channel scale. On the long epoch, whose post-stimulus mean now contains the salience and valence effects, the same pre-stimulus features reach 0.367 [0.342, 0.392] under z-scoring (permutation $p = 0.025$) and 0.325 [0.291, 0.358] under the fixed scale (paired difference $+0.042$, CI $[-0.001, +0.085]$). A linear readout thus classifies affect above chance from samples recorded before the picture appeared, through the normalization alone. The same normalization removes information where it belongs: on the long epoch, post-onset ERP-window features reach 0.447 under the fixed scale and 0.406 under z-scoring (paired -0.041 , CI $[-0.075, -0.007]$), because dividing each epoch by its own standard deviation discards the amplitude differences that carry the effect.

The window, more than the normalization, decides what

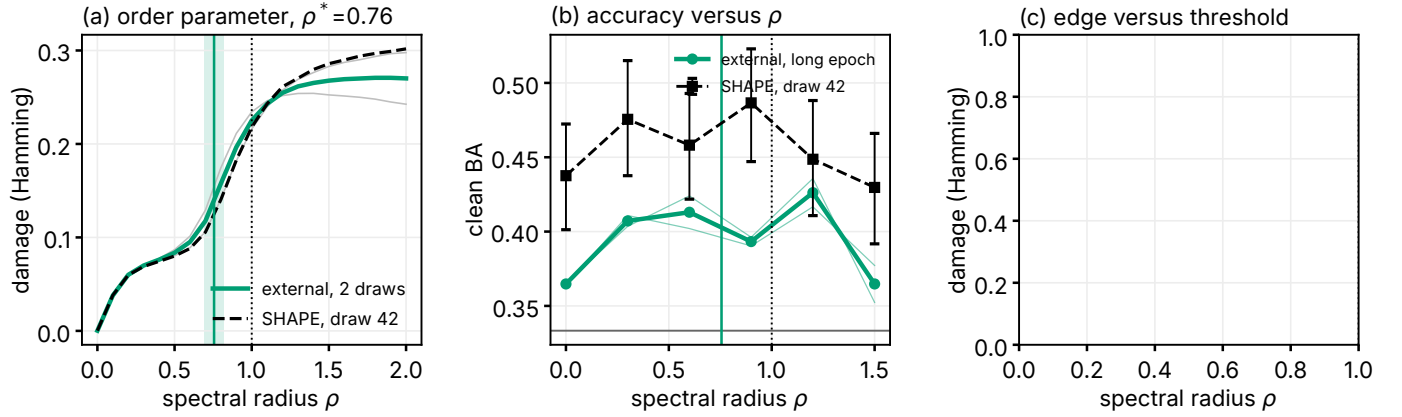


Fig. 5. The measured edge of stability. (a) Damage-spreading order parameter versus spectral radius for ten reservoir draws on the external drives (thin), their mean (thick), and SHAPE (draw 42, dashed); shading spans the transitions ρ^* of the ten draws, the solid line marks their mean, and the dotted line the echo-state heuristic $\rho = 1$. (b) Clean BA versus ρ for five draws on the external long epoch (thin) and their mean (thick), with SHAPE (draw 42, 95% CI); $\rho = 0$ removes the recurrence; chance = $1/3$. (c) Order parameter at three firing thresholds (draw 42); dots mark each ρ^* .

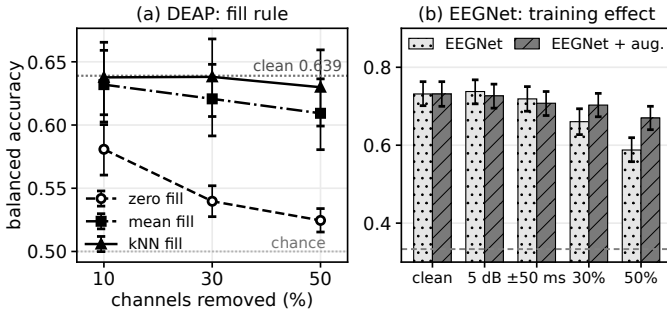


Fig. 6. Conference results [30]. (a) DEAP [28] (32 subjects, within-subject binary valence, band-power; CIs from resampling subjects): under channel removal, BA depends strongly on the fill rule. (b) SHAPE, EEGNet without and with channel-dropout and noise augmentation under the signal-level conditions of Table IV (subject-level 95% CIs): augmentation leaves clean accuracy unchanged and cuts the loss to electrode removal by more than half.

together with the above-chance margin it bounds how much the rule can matter; (iii) the normalization, and whether any window used for decoding precedes onset; (iv) trained-model results with and without augmentation; and (v) for reservoirs, the measured transition ρ^* and more than one weight draw.

Limitations. SHAPE could not be reanalyzed for this version, so the new experiments (origin sweep, pre-stimulus windows, spline repair, trial-count noise, ten reservoir draws) were run on the external cohort only, whose above-chance margin is small; effects that need headroom were therefore measurable on SHAPE and not on the external cohort, and only their mechanism could be tested on both. The external analysis used one pre-specified long epoch beyond the conference-matched one; no other windows were tried for accuracy. The reservoir's draws, thresholds, and leaks were varied only around the conference design, and EEGNet is the only trained architecture. DEAP was not reanalyzed. Clinical labels, graph topology, and structure–function coupling are outside the scope of this paper.

VII. CONCLUSION

CONCLUSION-PLACEHOLDER

ETHICS

The SHAPE study was collected by the Stony Brook University Laboratory for Clinical Affective Neuroscience under that laboratory's Institutional Review Board oversight. The external cohort and DEAP are publicly released datasets governed by their original collection protocols [28], [32], [33].

DATA AND CODE AVAILABILITY

For the SHAPE dataset, contact the Laboratory for Clinical Affective Neuroscience at Stony Brook University [29]. The external cohort is public under a CC BY-NC 4.0 licence [33], and DEAP is public [28]. The analysis code, which produces every number, table, and figure from these data, and the aggregate results are available from the authors.

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DRAFT: pending

Fig. 7. Signal-level perturbations on the external cohort (long epoch). (a) Recorded observation noise: subject-level BA (95% CI) when each test observation is re-averaged from 1, 4, or 16 of its single trials, and with all trials. (b) Electrodes removed as a zero trace (open) or repaired by spherical-spline interpolation (filled); dotted lines mark clean BA. Chance = 1/3.

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